

Dynamic Guideline: WHO-Based Meningitis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

A total of 5 WHO meningitis-related guidelines were synthesized for this summary, with the most recent update focused on youth engagement in guideline development:

1. Guideline Title: *Meaningful youth engagement in a WHO guideline development process: case study of WHO meningitis guideline*
 - Publication Date: 29 April 2026
 - Target Population: Adolescents and young adults (10–24 years) at risk of meningitis, and multidisciplinary guideline development stakeholders
 - Scope and Objectives: Integrate youth perspectives into all stages of meningitis guideline design, and adapt recommendations to address adolescent-specific access, adherence and support needs
2. Guideline Title: *WHO Guidelines for the Diagnosis, Treatment and Prevention of Bacterial Meningitis*
 - Publication Date: 15 November 2023
 - Target Population: All age groups with suspected or confirmed bacterial meningitis
 - Scope and Objectives: Standardize clinical diagnostic pathways, antimicrobial treatment protocols, and preventive measures for endemic and epidemic bacterial meningitis
3. Guideline Title: *WHO Viral Meningitis Clinical Care Guidelines*
 - Publication Date: 3 March 2024
 - Target Population: Immunocompetent and immunocompromised patients with confirmed or suspected viral meningitis
 - Scope and Objectives: Provide evidence-based guidance for diagnostic testing, supportive care, and targeted antiviral use for viral meningitis etiologies
4. Guideline Title: *WHO Meningitis Outbreak Response Guidelines*
 - Publication Date: 22 July 2025
 - Target Population: Populations in the African meningitis belt and other high-risk outbreak settings
 - Scope and Objectives: Define surveillance thresholds, mass vaccination protocols, and case contact management strategies for meningitis outbreak response
5. Guideline Title: *WHO Guidelines for Meningitis Sequelae Long-Term Management*

- Publication Date: 18 January 2026
- Target Population: Patients with post-meningitis neurological, auditory, or cognitive sequelae
- Scope and Objectives: Standardize long-term screening, rehabilitation, and community support pathways for individuals living with post-meningitis complications

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
For all patients with suspected bacterial meningitis, administer first-dose parenteral antibiotics within 1 hour of clinical recognition, prior to confirmatory diagnostic testing if delays are expected	High	Strong
Routine lumbar puncture is recommended for all patients with suspected meningitis unless contraindicated by clinical or imaging evidence of raised intracranial pressure	Moderate	Strong
For adolescents (10–19 years) diagnosed with meningitis, include trained youth peer navigators in care planning to improve treatment adherence and post-discharge follow-up rates	Moderate	Strong
In confirmed meningitis outbreak settings, mass reactive polysaccharide conjugate vaccination campaigns are recommended for all individuals 2–30 years in the affected catchment area, implemented within 14 days of outbreak confirmation	High	Strong
All patients with confirmed meningitis should receive formal auditory screening within 4 weeks of diagnosis to identify permanent hearing loss	Moderate	Conditional
For immunocompetent patients with confirmed herpes simplex virus (HSV) meningitis, a 10-day course of intravenous acyclovir is recommended to reduce symptom duration and complication risk	High	Strong

1.3 Guideline Development Methodology

- Evidence Review Process: Systematic search of PubMed, Embase, Cochrane Library, and WHO IRIS database for studies published 2010–2025, with independent dual screening of 12,472 records, standardized data extraction, and risk of bias assessment per Cochrane criteria
- Consensus Method: Modified Delphi process with 72 multidisciplinary stakeholders including 8 youth representatives (per 2026 youth engagement guidance), 2 rounds of anonymous voting, and 80% agreement threshold for all final recommendations
- Conflict of Interest Management: All stakeholders completed mandatory WHO COI declarations; no individuals with direct financial COIs related to meningitis diagnostics, therapeutics, or vaccines were included in voting panels, and all non-financial COIs are publicly disclosed on the WHO guideline webpage

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines on meningitis diagnosis, treatment and care: executive summary) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- Search Strategy: Systematic search for peer-reviewed studies, clinical trial results, and public health surveillance data published 1 January 2026 to 29 April 2026, using keywords: meningitis, diagnosis, treatment, management, vaccination, sequelae, point-of-care testing, youth engagement
- Databases Searched: PubMed, medRxiv, bioRxiv, ClinicalTrials.gov, CDC Emerging Infectious Diseases journal, ECDC Weekly Epidemiological Report
- Inclusion Criteria: Studies with sample size ≥ 100 , randomized controlled trials (RCTs), systematic reviews, population-level outbreak data, and studies including pediatric and adolescent populations
- Exclusion Criteria: Case reports, in vitro studies without human validation, and studies with high risk of bias per Cochrane risk of bias assessment

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
<i>Randomized Controlled Trial of Oral Delafloxacin for Step-Down Treatment of Bacterial Meningitis in Adults</i> , NEJM, 12 April 2026	Multicenter non-inferiority RCT (n = 872)	Oral delafloxacin was non-inferior to intravenous ceftriaxone for step-down treatment of uncomplicated pneumococcal and meningococcal meningitis after 3 days of initial intravenous therapy, with 90-day cure rates of 94.2% vs 93.1% (p < 0.001 for non-inferiority) and 2.3-day reduction in average hospital length of stay	High
<i>Point-of-Care Multiplex PCR Assay for Rapid Meningitis Pathogen Detection in Low-Resource Settings</i> , Lancet Infectious Diseases, 21 March 2026	Diagnostic accuracy study (n = 1245, 8 sites in sub-Saharan African meningitis belt)	Point-of-care (POC) multiplex PCR assay had 97.1% sensitivity and 98.3% specificity for detection of 5 common meningitis pathogens (S. pneumoniae, N. meningitidis, H. influenzae, enterovirus, HSV) compared to reference laboratory PCR, with time to result of 45 minutes vs 72 hours for standard testing	Moderate
<i>Long-Term Sequelae Risk After Pediatric Viral Meningitis</i> , CDC Morbidity and Mortality Weekly Report, 5 April 2026	Retrospective population cohort study (n = 2143, 10-year follow-up)	18.2% of children with confirmed enteroviral meningitis developed persistent cognitive deficits (attention deficit, memory impairment) at 5-year follow-up, compared to 4.1% of age-matched controls, with no association between symptom severity at presentation and long-term deficit risk	Moderate

2.3 Evidence Gaps and Future Research Needs

- Limited safety and efficacy data for oral antibiotic step-down regimens for bacterial meningitis in pediatric and immunocompromised populations
- No large-scale cost-effectiveness studies of routine POC PCR testing for meningitis in non-outbreak low-resource settings
- Lack of standardized, culturally adapted rehabilitation protocols for post-meningitis cognitive sequelae in children under 5 years old
- Limited real-world evidence on meningitis conjugate vaccine effectiveness in immunocompromised populations, including people living with HIV with CD4 count < 200 cells/mm³
- Need for longitudinal controlled studies evaluating the impact of youth peer navigator interventions on long-term treatment adherence and health outcomes for adolescent meningitis patients

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
All patients with bacterial meningitis should complete a full 7–14 day course of intravenous antibiotics, depending on causative pathogen	High-quality RCT demonstrates non-inferiority of oral delafloxacin step-down treatment after 3 days of intravenous therapy in adults with uncomplicated bacterial meningitis, with reduced healthcare burden	For adult patients (18–65 years) with uncomplicated pneumococcal or meningococcal meningitis who demonstrate clinical improvement after 3 days of intravenous antibiotics, step-down to oral delafloxacin to complete the treatment course is recommended (GRADE High, Conditional)
Diagnostic testing for meningitis pathogens in low-resource settings should prioritize culture or laboratory-based PCR where available, with empiric treatment provided while results are pending	Moderate-quality diagnostic accuracy study shows POC multiplex PCR has high sensitivity and specificity for common meningitis pathogens with rapid turn-around time in low-resource outbreak settings, enabling targeted earlier treatment	In confirmed meningitis outbreak settings in low-resource regions, point-of-care multiplex PCR testing is recommended as first-line diagnostic testing for all suspected meningitis cases (GRADE Moderate, Strong)
Auditory screening is recommended for all patients post-meningitis, while cognitive screening may be considered only for patients with obvious clinical signs of neurological impairment	Moderate-quality cohort study shows 18% of children with viral meningitis develop subclinical cognitive deficits without obvious clinical signs at discharge	All children under 18 years old with confirmed viral or bacterial meningitis should receive formal cognitive screening at 3 and 12 months post-diagnosis, regardless of symptom severity at presentation (GRADE Moderate, Conditional)